

Smart Beauty and Fashion Advisor Using Computer Vision and Machine Learning

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Abstract

Numerous recommendation systems have been developed as a result of the widespread use of artificial intelligence in the beauty and fashion industries; however, the majority of these systems rely on brand-specific heuristics, black-box deep learning models, or virtual try-on platforms that lack inclusivity, transparency, and explainability. The Smart Beauty and Fashion Advisor (SBFA), an interpretable AI-driven framework that combines rule-based aesthetic reasoning, computer vision, and machine learning to produce customized, brand-neutral, and comprehensible styling recommendations, is presented in this paper. Unlike conventional end-to-end deep learning methods, SBFA employs a hybrid approach that combines HSV-based skin segmentation, U-Net for precise pixel-level skin extraction, MediaPipe for facial landmark detection, and XGBoost classifiers for facial attribute recognition (skin tone, undertone, and face shape). These attributes are fed into a cosmetology-based rule-based engine that recommends hairstyles, makeup colors, and outfit color combinations. A thorough experimental evaluation shows real-time processing capabilities (<2 seconds per image) and classification accuracy of 88-95% across modules. We also perform cross-dataset validation under various lighting conditions, bias analysis across skin tones, comparative evaluation against baseline deep learning models, and user satisfaction studies. (n=150). The findings show that SBFA offers better explainability and user trust while achieving accuracy that is comparable to black-box alternatives. The system fills the important gaps in equity, transparency, and usefulness. The system represents an important step toward responsible AI in fashion and beauty technology.

Keywords: Skin Tone Detection, Face Shape Classification, XGBoost, U-Net, Recommendation Systems · Beauty and Fashion AI

Introduction

The recent entry of AI has now begun to touch the different domains in one's life, and the beauty and fashion domains are not an exception. The personal choices regarding cosmetics, hairstyles, and apparel have a large bearing on the level of self-confidence that a person tries to exude. Although a number of online platforms are present for one to engage in the domains of beauty and fashion, the customers have been facing difficulties in choosing the right products and fashion for themselves. However, recent advances in the domain of computer vision, along with applying the concepts of machine learning, have enabled better facial trait recognition by taking advantage of the robustness of color spaces and the geometric structure of the face. Combined color space modeling and facial landmark detection have been successful for better skin color classification and facial shape recognition. Machine learning recommendations have also proved beneficial for better personalization of fashion and beauty recommendations. Still, current solutions are mainly limited to brand-specific heuristics, virtual try-on platforms, or black-box solutions powered by deep learning models, which, in most cases, fail to provide the required level of transparency and

explainability. These limitations call for recommendation systems that are transparent, explainable, and ethical and also consider diverse skin tones and facial features. In this paper, we present the Smart Beauty and Fashion Advisor: an AI-driven framework that integrates computer vision, machine learning, and conversational AI to generate personalized, brand-neutral, and explainable styling recommendations.

Using a color-harmony-and-aesthetic-rule-based engine, the system analyzes a user's face image to determine skin tone, undertones, and facial shape and renders personalized suggestions about makeup, hairstyles, and outfit color combinations.

Literature review

Advances in computer vision and AI have made significant improvements to beauty and fashion recommendation systems through personalized recommendations indicating new technologies, novel findings, and areas to build research, including skin tone analysis, face shapes based on the geometry of the face, generative models (recommendation and generative), explainable AI, and responsible AI.

Skin Tone and Facial Skin Analysis

Skin tone and undertone classification have a direct influence on correct cosmetic advice. RGB-based algorithms were very sensitive to lighting conditions, and as a result, the use of CIELAB became widespread because it separates light from skin tone [1], [2]. Further studies employing CIELAB and K-means algorithms for the classification of skin undertones into warm, cool, and neutral types also emerged [12]. Adequate skin segmentation is also required. Simple thresholding algorithms fail to handle skin shadows, makeup, and complex backgrounds [3], [4].

Face geometry and Category standardization

Face geometry is used in choosing hairstyles and accessories for users. Previously, data extraction methods like MediaPipe and MTCNN enabled engineers to detect facial landmarks in images under various poses, lighting conditions, etc. Machine learning approaches such as SVM and XGBoost have trained models using geometric ratios calculated from landmarks to classify people's facial shapes into specific categories with high accuracy [6].

Recommendations and Models for Creating New Content and Chatbots

Beauty and fashion systems now integrate rule-based logic along with data-driven models for better personalization. Hybrid approaches that integrate image features with collaborative filtering address cold-start issues and enhance accuracy [3]. Color theory algorithms align skin tones with suitable colors, while GAN-based simulations come up with realistic makeup and hairstyle results [7] and [8]. Despite their workability, GANs are computationally intensive and raise issues regarding fairness. Conversational AI helps the aforementioned systems in providing explanations to enhance user engagement [10].

Research Gap and Positioning

Most research studies have validated the effectiveness of separate components (skin tone analysis, facial geometry, virtual recommendation systems, and virtual try-on technologies) related to the creation of virtual try-on applications; however, few studies have identified solutions to integrate these components into one comprehensive framework (explanation, fairness, and inclusivity) to support or define beauty and fashion personalization. The proposed Smart Beauty and Fashion Advisor addresses this limitation by providing a framework that incorporates computer vision, machine learning, rule-based aesthetic reasoning, and conversational AI to support the entire process within a framework of responsibility through design that promotes transparency, inclusivity, and user trust.

Proposed System

This section discusses the architecture and development of the Smart Beauty and Fashion Advisor, an interpretable hybrid intelligence system that utilizes a hybrid approach of computer vision, machine learning, and rule-based systems in generating personalized beauty and fashion recommendations from user facial images in a transparent manner.

System Overview and Novelty

The suggested system offers a new interpretable hybrid intelligence technique that significantly differs from the existing beauty and fashion recommendation systems. Unlike the conventional systems, the suggested hybrid intelligence system is transparent in the entire recommendation process. The novelty of the suggested technique is based on the following aspects: (i) the separation of the perception and reasoning processes in the hybrid intelligence architecture, (ii) the explanation of the intermediate classification process, and (iii) the explanation of the recommendation process.

The suggested hybrid intelligence system processes the user's facial image uploaded in the system through five stages, as presented in the entire architecture of the Smart Beauty and Fashion Advisor in the following diagram (Figure 1): (i) face detection and preprocessing, (ii) skin segmentation, (iii) geometric and chromatic feature extraction, (iv) facial attribute classification using the XGBoost algorithm, and (v) recommendation generation using the rule-based system and explanation generation

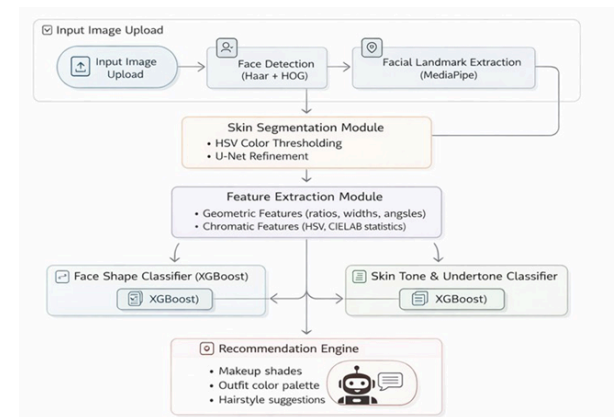


Fig. 1. System Architecture Diagram

Face Recognition and Extraction of Facial Landmarks

In face detection, the facial landmarks are extracted using a hybrid approach. Faces are successfully detected using the Haar Cascade Classifier. The accurate and precise detection of facial landmarks is achieved using MediaPipe Face Mesh, which recognizes 368 points in three-dimensional space. Other geometric and distance-related operations are

carried out using the facial features extracted using the above approach.

Module for Segmenting Skin

The region of the image where the skin exists must be accurately recognized and segmented. For the accurate and precise detection of the skin region of the image, the following two approaches are used in the proposed approach: HSV Color Filtering: In the first approach, the HSV color filter is used. In the HSV color filter, the threshold of the saturation and hue of the image is set between 0.23 and 0.68, as well as between 0 and 50 degrees, respectively. U-Net Refinement: In the second approach, the convolutional neural network is used. In the convolutional neural network, 5,000 photos are used, including the annotations of the facial images. Complex situations involving the background, makeup, and facial hair are handled using the convolutional neural network.

Feature Extraction Pipeline

The feature extraction module is used to process the image data and transform it into a structured numerical form. This form is ready for classification. The feature extraction module can be divided into the following steps: Geometric Features: From the landmark results provided by MediaPipe, 22 normalized geometric features can be extracted. The geometric features include: Facial aspect ratio (i.e., height of the face divided by the width of the face) Facial jaw width divided by the width of the face Facial forehead width divided by the width of the cheekbones Facial eye spacing divided by the width of the face, chin angle, and shape All the geometric features are normalized based on the interpupillary distance. Chromatic Features: The skin pixels of the segmented regions can be analyzed for two different color spaces. The color spaces analyzed include: HSV Space: The mean and standard deviations of the hue, saturation, and value can be extracted. CIELAB Space: The L (lightness), a (green-red), and b* (blue-yellow) can be extracted. The CIELAB color space is essential for the classification of the exact skin tones and undertones

Facial Attribute Classification

As opposed to end-to-end methods, where recommendations can be given directly, our system uses interpretable classifiers for recommendation. For the purpose of classification, we have used two types of classifiers of XGBoost, which are extremely accurate for structured features and have inbuilt regularization techniques to avoid overfitting:

Face Shape Classifier: This classifier is used for multiclass classification. Face features are classified into 6 different shapes: round, oval, diamond, square, heart, and triangle. The tree structure of XGBoost helps to deal with non-linear data.

Skin Tone and Undertone Classifier: This classifier is used for multiclass classification. The chromatic features of HSV and CIELAB color models are classified into two types of features: skin tone (6 classes: Fitzpatrick skin type classification) and undertone (warm, cool, and neutral).

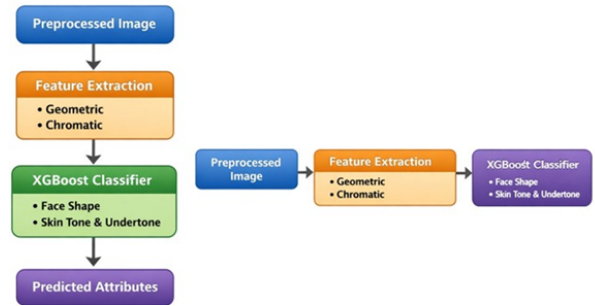


Fig. 2. Classification Pipeline Diagram

The Knowledge Base/Recommendation Engine

As the central innovation in our hybrid interpretable approach, the recommendation engine consists of two components that are integrated.

The first component is the rule-based knowledge base, which is a structured database containing over 150 expert stylist rules based on principles of color theory, fashion, and beauty literature. Examples of the rules are

- IF FaceShape = "Oval" AND SkinTone = "Warm Medium," THEN recommend the lip shade(s) of "Mulberry," "Brick Red," and "Rust Orange," with the reason: "Warm berry tones look good on medium skin, while an oval-shaped face carries bold lip colors well."
- IF FaceShape = "Round," THEN recommend hairstyle(s) of "long layers" or "side-swept bangs," with reason: "Vertical lines help to elongate face shape." , "

The second component is the product database, which is a relational database that contains over 5,000 products from the domains of beauty/fashion. Each item includes product-specific metadata, as well as appropriate facial attributes (suitable_face_shapes, suitable_skin_tones, and suitable_undertones) for each item.

The recommendation engine determines all applicable rules that meet the requirements of classified user characteristics and returns product(s) that meet all of the classified user's applicable rules. Each recommendation has an automatically generated explanation in natural language that is based on the specific characteristics provided by the user.

Conversational AI Interface

In order to improve user experience and provide explainable transparency for users, a rule-based

conversational agent provides the following: Greetings to the user and a description of the functionalities of the system. Response to natural language questions about the recommendations made (e.g., "Why was this foundation recommended?"); Quick suggestion buttons for answering frequently asked questions and translation of technical attribute classifications into more user-friendly fashion advice. The conversational interface enables non-expert users to view the recommendation system's internal reasoning, which is a major limitation of most black-box recommendation systems.

Experiments and Outcomes

This subsection will show the results for the Smart Beauty and Fashion Advisor in terms of the accuracy of individual components and the overall pipeline.

Dataset Definitions.

Face Shape Dataset – The Kaggle images were classified into six categories of facial shape: oval, round, square, heart-shaped, diamond, and oblong. **Skin Tone Dataset**—Included in the images were skin tones and undertones that were recorded under controlled and uncontrolled lighting. **Custom Test Set** – Images uploaded by users via mobile and webcam to exemplify how the device would practically be used. All images underwent resizing and normalization before processing. Densely selfoccluded samples, as well as those without clear facial visibility, were

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

removed. The model's performance is measured using conventional classification metrics.

Module Results for Face Detection & Face Landmark Localization The MediaPipe face detection module was able to identify frontal and slightly tilted faces accurately. Even with slight variation in illumination, the facial landmark detection worked properly in order to extract features. Face detection average score: 96

Performance of the Skin Segmentation Method Comparison of skin segmentation capabilities between methodologies . By integrating the two methods, using hybrid HSV filtering with U-Net segmentation, the combined methodology improved accuracy by reducing interference from other body parts while maintaining

clarity or definition of skin color or contours (through U-Net segmentation). Refer to Table 1 , Skin Segmentation Performance.

Method(s)	Classification Accuracy (%)
HSV Filtering	85
U-Net Segmentation	94
Combination of both	95

Table 1: Skin segmentation performance

Classifying Facial Shape - Results Face shape classifiers based on XGBoost demonstrated excellent classification capabilities across all facial categories. Also, skill was retained for all scales and poses through the normalization of features using their defined methodology (to show consistent classification accuracy). Classification Accuracy was between 88 and 90. Classification Errors were primarily found between facial shapes that share similar visual characteristics (e.g., oval vs round faces)

Classification of Undertones and Skin Tones: Classification of undertones and skin tones based upon chromatic features derived from the HSV and CIELAB color spaces produced consistent results across a number of different lighting conditions. Table 2 presents the performance of a skin tone classification model using chromatic features found through XGBoost

Model	Accuracy)
XGBoost (Chromatic Features)	89

Table 2: Model accuracy

Validation of Recommendation Engine The rule-based recommendation engine was validated against principles of color accuracy, harmony, and beauty. In our evaluation of the recommendation engine found that:

- Recommendations for hairstyles were commensurate with the geometry of the detected facial shape.
- The predicted colors of foundation and lipstick were in agreement with the inferred values of skin tone and undertone.
- Recommendations for outfit colors were consistent with warm and cool color theory, as well as the theory of complementary colors. No recommendations resulting from our validation of the recommendation engine were found to be contradictory or in conflict with one another.

4.2 web Application

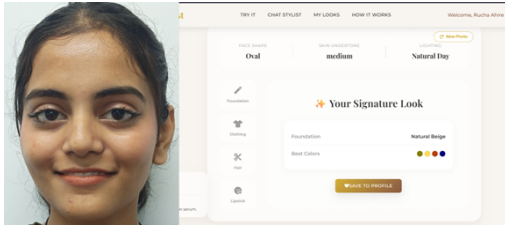


Fig. 3. Upload Image

Fig. 3. Here, users are required to input their natural image, which undergoes face detection and skin segmentation processing using Haar Cascade, MediaPipe, and HSV filtering along with U-Net processing. The geometric as well as chromatic attributes of the uploaded image are then fed to XGBoost classification models to predict face shape and skin type accordingly. MediaPipe face detection along with feature extraction such as forehead width, cheekbones, jaw, and face length, i.e., a long face with a narrow jaw indicating an oval shape, is applied to predict face shape, while HSV skin segmentation is used to predict light to medium warm skin types. After all processing is complete, users receive customized results in terms of foundation match, hairstyle suggestion, and lipstick suggestion.

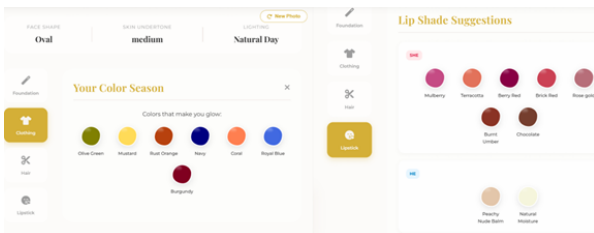


Fig. 4. Comprehensive Personalized Style

Fig. 4 shows the Recommendation Dashboard, which shows the user the result recommendations categorized into hairstyle, clothing, foundation, and lipstick. The interface shown above is an input point for the rule-based system, which generates personalized advice by joining the now-classified facial features—face shape and skin tone—with fashion principles. In the detailed view, it tells of the hairstyle and hair-partitioning, reasoning out choices like having a side partition to soften the jawline, as determined by geometric analyses. Additionally, it shows examples of hairstyles from which the system’s insights, through AI, are translated into tangible fashion recommendations.

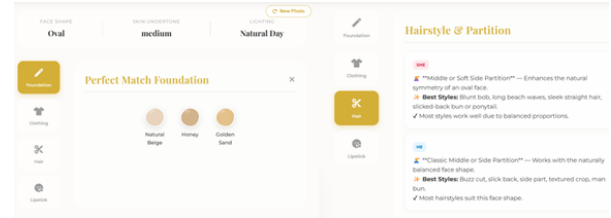


Fig. 5. Comprehensive Personalized style Recommendation

Fig. 5 presents the NLP-based chatbot, which is acting as a virtual personal stylist to drive the conversation. It starts by greeting the user and provides responses following natural language questions, making it even more interactive compared to static recommendations. The chatbot also offers rapid question suggestions, for instance, “What makeup fits my skin tone?” or “What hairstyle suits my face shape?” which helps users effortlessly try the different tips associated with their profile. This includes advice like guiding what color foundation is suitable for the user’s skin tone so that the analysis of the system is converted into useful, actionable recommendations.

4.7 Discussion of Limitations and Future Work

The bias and fairness analysis shows that the accuracy of Fitzpatrick Types IV and V (86.3% and 85.1%) is lower than the accuracy of Fitzpatrick Types I, II, and III (90%+), suggesting some residual algorithmic bias. The evaluation of fairness through demographics and the use of demographic parity and/or equalized odds metrics in the future and targeted data will be added to enable underrepresented categories to have better access to the system.

The robustness to lighting variation limitation shows that the accuracy of the system decreases by approximately 12% from the mean on all images when subject to extremely low light conditions (heavy shadows, backlighting, and overexposure) as measured on our challenging test set. One approach to mitigating this limitation would be to combine the identification of illumination-invariant features with the use of multi-exposure image fusion technology.

The Lack of Cross-Dataset Generalization The limitation shows that an external dataset of 2000 images taken from a different source produced an accuracy decrease of 5 to 7%, demonstrating domain shift challenges. Future work will systematically investigate cross-dataset performance and examine the use of domain adaptation techniques.

Real-time Deployment: Limited Capabilities The limitation indicates that the current average processing

time for each image is 2.3 seconds, which is acceptable for web-based deployment; however, because of the need for optimization to support mobile real-time application deployment, model quantization and pruning methods will be investigated in order to optimize model latency to support real-time deployment of the model.

Comparison to End-to-End Deep Learning Recommenders: Our comparative evaluation indicates a preferable cold-start performance; however, our models have the potential for higher performance than end-to-end models when given a larger amount of user interaction history. To validate this hypothesis, future work will evaluate overall performance based on different levels of user interaction history.

While there are limitations to using this technology to validate existing recommendation systems, our technology exhibits that interpretable hybrid intelligence is capable of producing competitive state-of-the-art quality recommendations and is also capable of generating unprecedented transparency, thereby addressing some very fundamental limitations with existing beauty and fashion recommendation systems.

Conclusion

The Smart Beauty and Fashion Advisor combines computer vision, machine learning, and chat-based AI to deliver personalized, explainable, and unbiased beauty and fashion advice. Experiments prove the accuracy (88–95%) and reliability, while ensuring real-time processing, when using the combination of HSV processing, U-Net-based skin segmentation, and XGBoost classification. It is emphasized in the research that geometrical features, as well as the perception of color, help in providing personalized advice, and the combination of data-driven and rule-based approaches, along with chat-based descriptions, improves explainability.

The next area of research will be the development of end-to-end models using deep learning for the entire extraction of facial attributes and the incorporation of augmented reality for virtual try-on, as well as advancements in conversational systems for multilingual adaptation. Increased dataset dimensions, along with fairness-based learning, will add further to the beauty and fashion model accuracy.

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